

UAE HACKATHON 2026 · PIONEERS

KAMouflage Vision

AI that finds UAE wildlife in camera-trap photos, so experts only check what matters.

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MOCCAE wildlife monitoring challenge

Our team

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The problem

MOCCAE wildlife monitoring challenge

Camera traps across UAE reserves take thousands of photos. Experts check every one by hand, so verified wildlife records arrive too late to guide conservation decisions.

01

Too many photos

Most frames are empty, triggered by wind or heat, yet someone still opens each one.

02

Hard to read

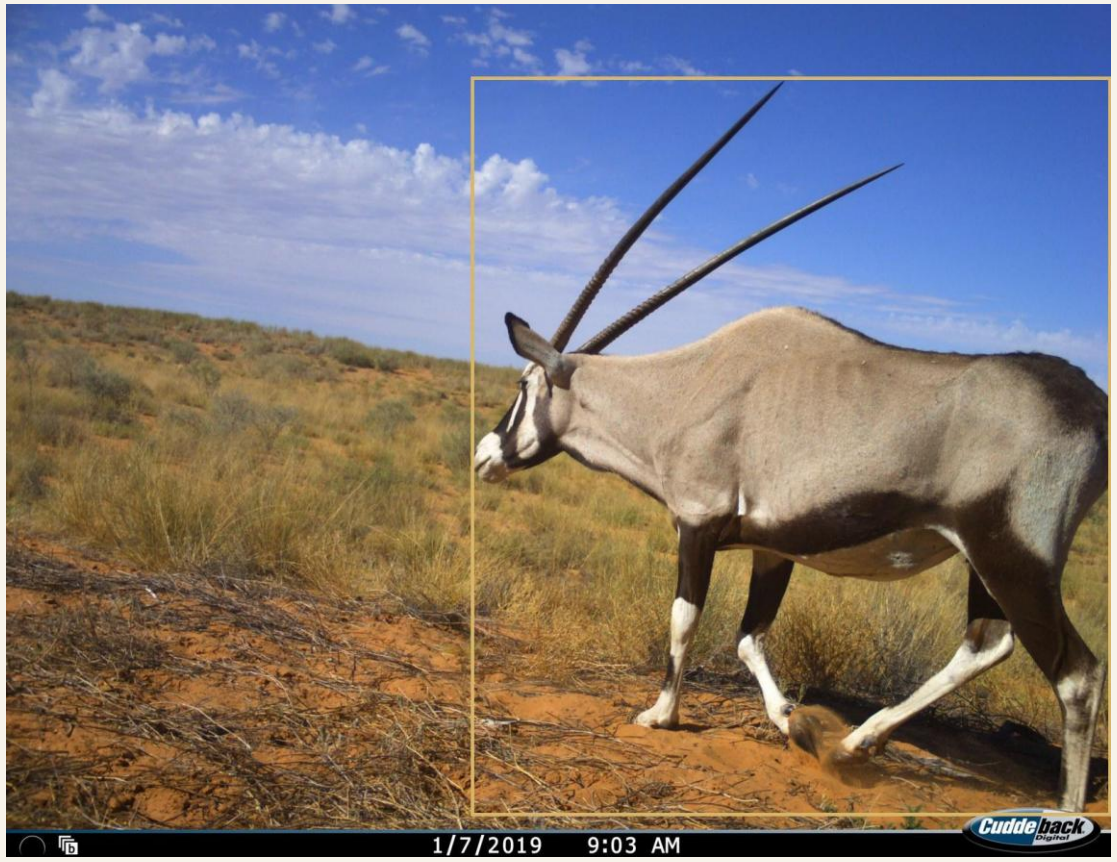
Night and infrared shots, partial animals and look-alike species slow experts down.

03

Insights come late

Records reach the map long after the photos were taken, and coverage gaps stay hidden.

Our solution: KAMouflage Vision



Detect

Find animals in a batch of camera-trap images.

Suggest

Identify a species and combine burst evidence.

Verify

A reviewer confirms or corrects the event.

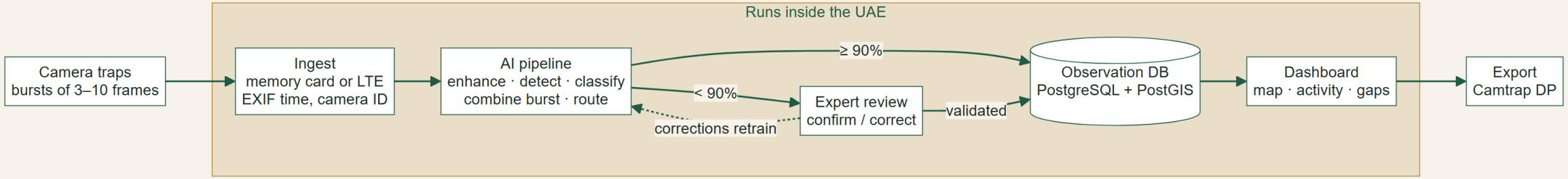
Understand

Confirmed events feed the map and activity chart.

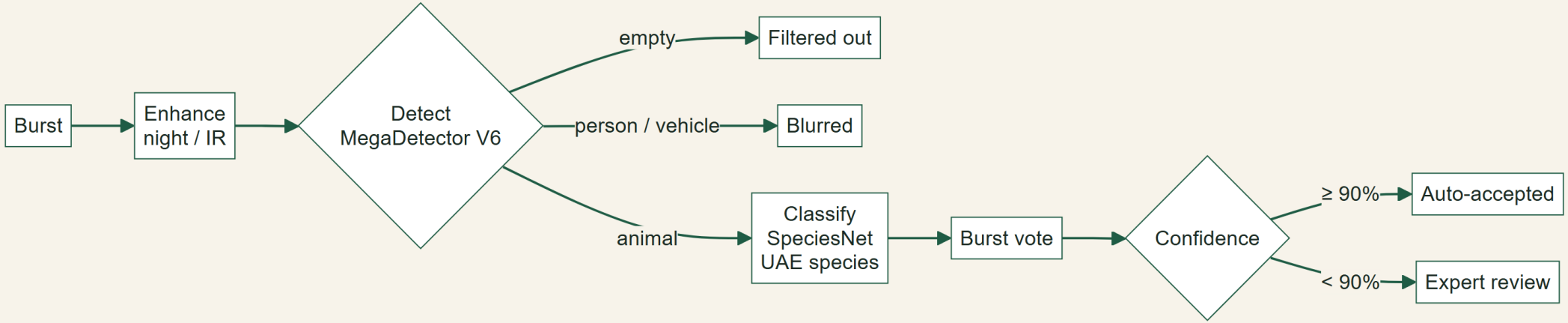
Real inference example: gemsbok, public southern-African proxy image.

How it works

SYSTEM

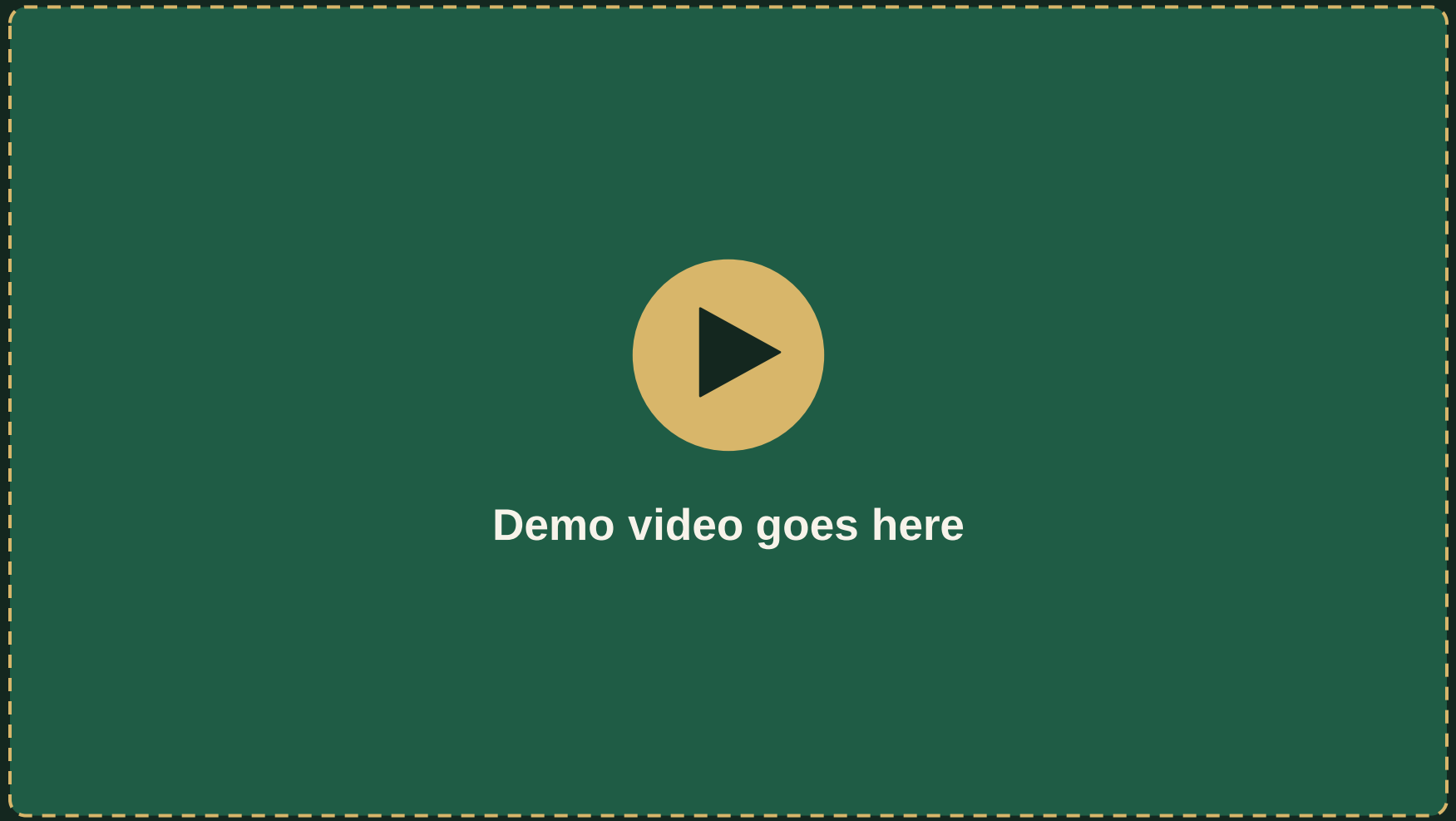


ONE BURST OF PHOTOS



All models run locally and are MIT / Apache-2.0 licensed. No images leave the UAE.

Live demo



Tested on 600 camera-trap photos

276 bursts with known labels, run through our detector and classifier

86.8%

Animal bursts kept

165 / 190

67.4%

Empty bursts filtered

58 / 86

48.9%

Species named exactly

93 / 190

Empty photos never reach an expert. Unsure species go to review, not into the records.

UAE context: 44 MOCCAE reserves and 381 bird species via Bayanat, plus GBIF species records.

Public southern-African proxy photos (LILA BC); UAE camera sites simulated. UAE field accuracy needs permitted local data.

Building on what exists

Wildlife Insights	AddaxAI	KAMouflage Vision
Cloud platform for camera-trap management, AI identification and human validation.	Open-source desktop tool. Models run locally, offline after setup.	Built for MOCCAE: UAE government data, expert review queue, AI Advisor, Camtrap DP export. Runs inside the UAE.

Our advantage

A workflow tailored to MOCCAE, with government data integration and a path to authority-hosted deployment.

Sources: wildlifeinsights.org; docs.addaxai.com

Business Model Canvas

How we deliver	The value	How adoption works
Partners Authority sponsor, reserve team and ecologists.	**Value** Less repetitive review and traceable observations.	**Customers** Monitoring staff use it. Conservation authorities sponsor it.
Activities Inference, verification, evaluation and reporting.	**Relationships** Reviewer onboarding, support and quality reviews.	**Channels** A challenge-owner pilot and government integration.
Resources Permitted images, expert labels, software and hosting.	**Costs** Engineering, review labor, compute, storage and security.	**Funding** Sponsored pilot, then support or integration contracts.

The next 90 days

01-30

Establish the baseline

Prepare local GPU, database and storage. Agree access controls and a local test set.

31-60

Improve local quality

Fine-tune SpeciesNet on verified UAE crops. Calibrate and test held-out sites.

61-90

Test real operations

Review new retrievals. Audit errors and compare human time. Make a scale decision.

The ask: one site, permitted images and expert reviewers.

Targets: 99% animal retention, 95% accepted precision, 50% less review time.

Runs on existing authority hardware; budget scoped with the authority's IT team. All images and models stay on authority-controlled infrastructure.

Quick facts: how the AI names an animal

MegaDetector V6

Is anything there?

Draws a box around an animal, person or vehicle. It does not name the species.

Microsoft · MIT

SpeciesNet

Which animal is it?

Names the animal in the box, e.g. "Arabian oryx, 94%". Trained on about 65 million camera-trap photos and knows about 2,000 labels.

Google · Apache-2.0

UAE geofence

Could it live here?

We tell SpeciesNet the photos come from the UAE, so it never suggests animals that don't live here.

Country filter

HOW A DECISION IS MADE

Burst vote
3–10 frames per trigger are combined

90% or more
Accepted automatically

Below 90%
An expert confirms or corrects

Empty photos
Filtered out before anyone sees them

Night and people
Night shots enhanced, people blurred

Quick facts: data and product

Data

- 600 test photos (276 bursts) from LILA BC, a public camera-trap library
- UAE protected areas and bird list from MOCCAE via Bayanat
- Species records and IUCN Red List status from GBIF
- UAE camera positions simulated until real photos are shared

Product

- Python + Streamlit app
- Import, gallery and review queue
- Dashboard: map, activity by hour, coverage gaps
- AI Advisor: missing cameras, clock errors, threatened species
- Camtrap DP export, the format GBIF uses

Why these models

- Free and permitted for government use (MIT / Apache-2.0)
- Avoided MegaDetector v5: its AGPL license would force everything built on it to be open-sourced
- Everything runs locally: no photos go to a cloud service

How we built it

Area	What we did
Challenge fit	Built around MOCCAE's workflow: detect, suggest, expert verify, insights.
Data	MOCCAE reserves and bird list (Bayanat), GBIF species and IUCN status, 600 LILA BC camera-trap photos.
AI	MegaDetector V6 (MIT) and SpeciesNet (Apache-2.0), burst voting, confidence routing, AI Advisor.
Product	Working app: import, gallery, review queue, insights dashboard, AI Advisor, Camtrap DP export.
Feasibility	Runs locally; 90-day pilot plan; data stays on authority infrastructure.
Market	Clear buyer (conservation authorities), pilot-first adoption, support and integration contracts.

Sources: MOCCAE/FCSC via Bayanat; GBIF.org; LILA BC Snapshot Karoo/Kgalagadi (CDLA-Permissive); microsoft/MegaDetector; google/cameratrapai; camtrap-dp.tdwg.org

What works today, what the pilot adds

Implemented MVP	Funded pilot work
MIT detector and Apache SpeciesNet classifier. Species names, documented footer exclusion for proxy images, resumable caches.	Local taxonomy coverage, calibrated thresholds and independent tests on unseen UAE sites.
Review decisions saved, reviewer-confirmed analytics, CSV export in Camtrap DP format.	Authenticated access, protected-location policies, monitoring, backup and validated Camtrap DP packages.
Local processing and cached demo. No cloud inference API required.	Authority-approved UAE deployment, security review, operational ownership and support agreements.

Production hosting and data reuse require authority approval.

Evaluation protocol and limitations

Unit of evaluation	276 complete source bursts. Labels are per burst, so no frame-level or animal-count claim.
Exact species scoring	Case-insensitive name aliases. Any label can match in a multi-label burst. Genus-level answers count as wrong.
Data limitations	Curated class balance, few rare-class samples, unknown overlap with pretrained training data. No UAE generalization claim.
Risk control	Bursts below 90% confidence go to an expert. Corrections become training labels only after quality checks.
Next experiment	A site-separated UAE test set and an assisted-vs-manual time study with monitoring staff.



Thank you

Questions?

github.com/abdalla980/KAMouflageVision-UAE-HACKATHON26

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